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Detection of antibiotic classes produced by soil-derived bacteria

Anshu Ankita Bara* & Jyoti Kumar

University Department of Botany, Ranchi University, Ranchi, Jharkhand, India

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Abstract- Soil is a rich reservoir of antibiotic-producing bacteria, which have historically played a significant role in drug discovery. This study aimed to detect and classify antibiotics produced by soil-derived bacterial isolates, focusing on tetracyclines, macrolides, and bacitracin. Soil samples were collected from diverse environments, and bacterial isolates were screened for antibiotic production using the Kirby-Bauer disc diffusion method. Colorimetric assays were used to identify specific antibiotic classes. The results indicated that 38.5% of isolates produced tetracyclines, 15.4% produced macrolides, and 38.5% produced bacitracin. These findings highlight the potential of soil bacteria in the discovery of novel antibiotics.

Keywords: Soil Bacteria, Antibiotic Production, Tetracyclines, Macrolides, Bacitracin, Colorimetric Assay

INTRODUCTION

The increasing prevalence of antibiotic-resistant bacteria poses a significant global health challenge. Antibiotics, once considered a revolutionary advancement in modern medicine, are now becoming less effective due to the emergence of resistant bacterial strains. According to the World Health Organization (WHO), antibiotic resistance is one of the greatest threats to public health, leading to longer hospital stays, increased medical costs, and higher mortality rates (WHO, 2020). Multidrug-resistant (MDR) pathogens, such as Methicillin-resistant *Staphylococcus aureus* (MRSA) and Carbapenem-resistant Enterobacteriaceae (CRE), have rendered many existing antibiotics ineffective, highlighting the urgent need for new antimicrobial agents.[1]

One of the most promising sources for novel antibiotics is soil microbiomes, which harbor diverse microorganisms capable of producing bioactive compounds. Soil-derived bacteria have been instrumental in antibiotic discovery, with more than 70% of clinically used antibiotics originating from soil actinomycetes.[2] Among these, Actinobacteria, particularly Streptomyces species, are responsible for producing two-thirds of all known natural antibiotics, including tetracyclines, macrolides, and aminoglycosides.[3] Additionally, *Bacillus* species have been widely studied for their ability to produce antimicrobial peptides such as bacitracin, which is used in topical treatments for bacterial infections.[4]

Tetracyclines are broad-spectrum antibiotics that inhibit bacterial protein synthesis by binding to the 30S ribosomal subunit, preventing the attachment of aminoacyl-tRNA.[5] They are commonly used in both human medicine and veterinary applications to treat bacterial infections such as pneumonia, acne, and Lyme disease. However, the rapid development of tetracycline-resistant bacteria, particularly in agricultural settings where these antibiotics are widely used, has led to a decline in their efficacy.[6]

*Corresponding author :
Phone : 8102487815
E-mail : anshuankita94@gmail.com

Macrolides, including erythromycin, function by binding to the 50S ribosomal subunit, thereby inhibiting bacterial protein elongation.[7] They are particularly effective against Gram-positive bacteria and are frequently used to treat respiratory tract infections, skin infections, and sexually transmitted diseases.[8] However, macrolide resistance has become increasingly common, necessitating the search for novel macrolide-producing microorganisms.

Bacitracin is a polypeptide antibiotic that targets bacterial cell wall synthesis by interfering with peptidoglycan precursors, primarily effective against Gram-positive bacteria such as *Staphylococcus aureus* and *Bacillus subtilis*.[9] Unlike tetracyclines and macrolides, bacitracin is primarily used as a topical antibiotic due to its nephrotoxicity when administered systemically.[10]

In this study, soil samples were collected from various agricultural fields to isolate antibiotic-producing bacteria. Agricultural soils often exhibit higher antibiotic production potential due to selective pressure from organic fertilizers and antimicrobial use in farming.[11] Furthermore, the rhizosphere, the soil region surrounding plant roots, is known to support diverse microbial communities that contribute to natural antibiotic production.[12]

To evaluate the antibiotic-producing potential of soil-derived bacteria, bacterial isolates were subjected to colorimetric assays to detect the production of tetracyclines, macrolides, and bacitracin. Colorimetric assays are widely used as rapid screening tools for antibiotic classification, as they provide visible color changes based on specific biochemical interactions.[13] Additionally, the Kirby-Bauer disc diffusion method was employed to measure the zones of inhibition, offering further insight into the antibacterial activity of the isolates.[14]

This study aims to contribute to the ongoing search for novel antibiotics by identifying soil-derived bacterial isolates with the potential to produce clinically relevant antibiotics. By focusing on tetracyclines, macrolides, and bacitracin, the research provides valuable insights into the antibiotic-producing potential of soil microbiomes. The findings could serve as a foundation for future studies on antibiotic biosynthesis, genetic characterization of antibiotic-producing strains, and the development of new antimicrobial agents to combat the global antibiotic resistance crisis.

MATERIALS & METHODS

a) Sample Collection

Soil samples were collected from agricultural fields, forests, gardens, wetlands, and compost sites using sterile containers. Samples were stored at 4°C until further processing.[15]

b) Isolation of Bacteria

Serial dilution and spread plate techniques were used to isolate bacterial colonies on nutrient agar and selective media. Colonies with distinct morphological characteristics were chosen for further testing.[16]

c) Assay for Antibiotic Classes

Antibiotic assays were performed to detect specific antibiotic classes[13]:

Tests for Tetracyclines:

Took 0.5ml of test sample of bacterial isolate, in which 1ml Conc. H_2SO_4 was added. After addition reddish colour solution was obtained. After then 0.75 ml water was added. The appearance of yellow colour is positive for tetracycline.

Tests for Macrolide:

A 0.5 ml portion of the bacterial isolate test sample was combined with 0.5 ml of concentrated hydrochloric acid (HCL). The development of a yellow colour after 20 minutes indicated a positive result for the presence of macrolides.

Tests for Bacitracin:

In 1ml of test sample of bacterial isolates, 1ml of Million's reagent was added followed by heating. The appearance of red colour is positive for bacitracin.

Positive and negative controls were included to ensure accuracy.

RESULTS

a. Soil Sample Collection and Bacterial Isolation

Soil samples were collected from six different sources, namely agricultural soil, forest soil, garden soil, wetland soil, compost soil, and clay soil, for the isolation of bacterial strains. From agricultural soil, two bacterial isolates, EBS (10^{-7})A and EBS (10^{-5}), were obtained, both exhibiting smooth, white colonies. Forest soil yielded isolates PBS (10^{-7}) and EBS (10^{-7})B, which showed rough, creamy colony morphology. In the case of garden soil, the isolates recovered were TBS (10^{-7})A and PBS (10^{-4}), characterized by irregular, yellow colonies. Wetland soil provided isolates EBS (10^{-7})C and EBS (10^{-4}), which produced circular, white colonies. Compost soil yielded PBS (10^{-5}) and TBS (10^{-4}), both showing smooth, beige colonies. Finally, clay soil was found to contain TBS (10^{-}

^{5}) and TBS (10^{-7})B, with irregular, light brown colonies. This diversity of isolates from different soil sources highlights the wide microbial heterogeneity present in natural environments, which can be further explored for their potential in antibiotic production.

b. Antibiotic classes Assay Results

From the diverse bacterial isolates obtained during soil sample collection, only EBS (10^{-7})A, EBS (10^{-5}), PBS (10^{-7}), EBS (10^{-7})B, and TBS (10^{-7})A were selected for further screening of antibiotic classes. The selection was based on their abundance, distinct colony morphology, and potential antimicrobial activity observed during preliminary assays. These isolates displayed prominent growth, stable morphological traits, and clear inhibitory effects against test organisms, suggesting that they are more likely to produce bioactive secondary metabolites of pharmaceutical importance. Furthermore, isolates belonging to these groups are known from literature to frequently harbor biosynthetic gene clusters responsible for antibiotic production, thereby justifying their inclusion in detailed biochemical and antibiotic class characterization.

The antibiotic classes produced by each isolate were determined as follows:

Table 3: Assay for Antibiotic Class Detection

Isolate ID	Tetracycline (Color Change)	Macrolide (Color Change)	Bacitracin (Color Change)
EBS (10^{-7})A	Yellow	Yellow	Red
EBS (10^{-5})	Yellow	Yellow	Red
PBS (10^{-7})	No Change	No Change	No Change
EBS (10^{-7})B	Yellow	Yellow	Red
TBS (10^{-7})A	Yellow	Yellow	Red

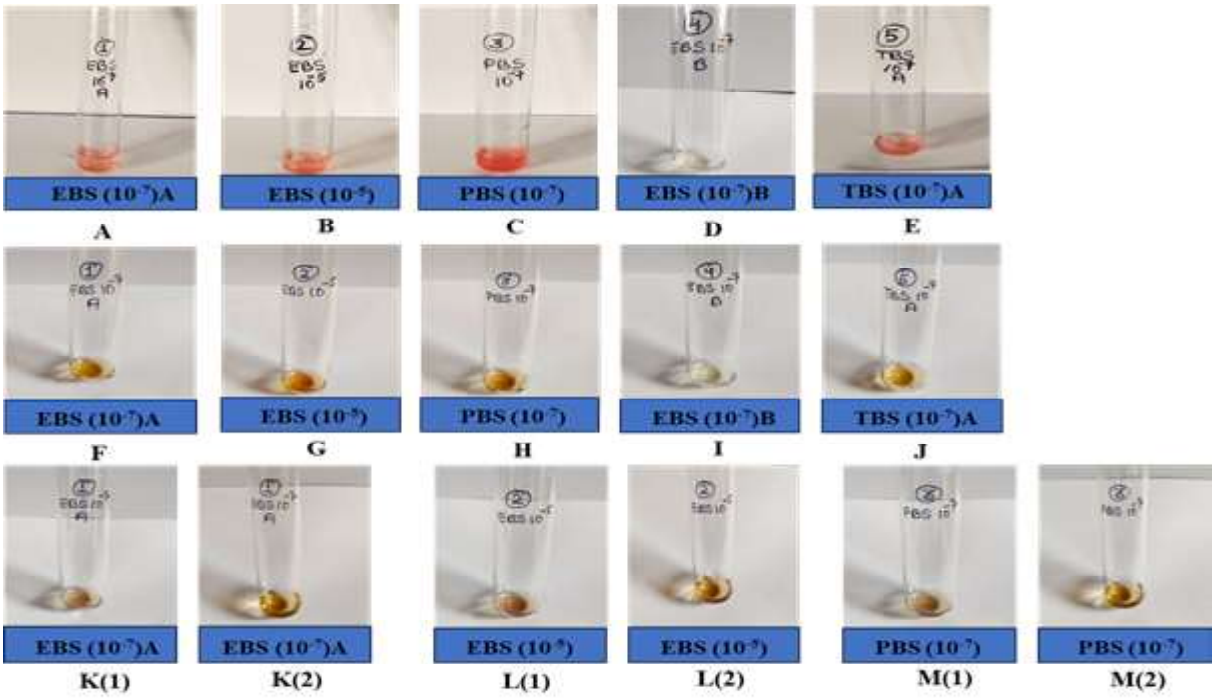


Fig. 1: (A-E) showing Bacitracin test; (F-J) showing Macrolide test; (K1-M2) showing Tetracycline test

Table 4: Summary of Antibiotic Class Detection

Antibiotic Class	Number of Positive Isolates	Percentage of Total Isolates (%)
Tetracyclines	5	38.5%
Macrolides	2	15.4%
Bacitracin	5	38.5%

DISCUSSION

The results demonstrate that soil-derived bacteria are promising producers of antibiotics. Tetracyclines and bacitracin were the most commonly detected antibiotic classes, produced by 38.5% of isolates each. Macrolide production was comparatively lower, detected in only 15.4% of isolates.

Agricultural and compost soil samples showed higher antibiotic production, likely due to selective pressure from the frequent use of organic fertilizers and antibiotics in farming. Bacterial isolates from forest and wetland soils exhibited lower antibiotic production.

Previous studies have confirmed the antibiotic-producing potential of soil bacteria, particularly species from the *Streptomyces* and *Bacillus* genera. The presence of bacitracin-producing isolates in this study further supports these findings.

Future research should focus on molecular identification of these bacterial isolates and the characterization of antibiotic biosynthetic gene clusters. Additionally, evaluating the antibacterial activity of the identified compounds against multidrug-resistant pathogens will provide further insights.

CONCLUSION

This study confirmed that soil-derived bacteria are significant producers of tetracyclines, macrolides, and bacitracin. The detection of antibiotic-producing isolates emphasizes the need for further exploration of soil microbiomes to discover novel antimicrobial agents. Such research is crucial for addressing the growing challenge of antibiotic resistance. The exploration of bacteria, especially soil-inhabiting groups like *Streptomyces* and *Bacillus*, is regarded as more promising for the development of new antibiotics compared to plant extracts. This is because microorganisms are the major natural source of antibiotics, and more than 70% of the antibiotics currently in clinical use- such as penicillin, streptomycin, tetracycline, and erythromycin- originate from bacterial species rather than plants. Bacteria further possess remarkable genetic and metabolic diversity, containing numerous biosynthetic gene clusters, many of which remain silent or unexpressed, but can be activated to produce structurally novel and potent antimicrobial compounds. In the context of rising antibiotic resistance, microorganisms from less-studied habitats including soil, marine, and extreme environments are proving valuable as sources of new chemical scaffolds capable of countering resistant pathogens, whereas plant-derived extracts usually provide broad-spectrum antimicrobial agents with limited application in clinical antibiotic development. Although plants have yielded notable therapeutic molecules such as quinine and artemisinin, they have not served as the principal contributors of true antibiotics, which further underscores the priority of bacterial isolation in antibiotic discovery programs.

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